VITyarthi | Open Source Software

The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5
Max Marks: 100

Student Name	Registration Number
ARPIT RAGHAV	24BAI10598
Slot	**Date of Submission**

OPEN SOURCE SOFTWARE ANALYSIS REPORT — PYTHON

Introduction

- Open source software has really changed how we make and use software over the world. It is different from software that is owned by a company. Open source software lets people look at the code make changes and share it with others. This is good because it helps people work together and come up with ideas.
- This report is about **Python**, a popular open source programming language. **Python** is used for things like making websites, artificial intelligence looking at data, machine learning and automating tasks. **Python** is easy to understand and use so people who are just starting out and experienced developers all like it.
- The report talks about how **Python** was made what it is about how it is licensed and why it is important, in the Linux world. It also compares open source software to software that is owned by a company. We even. Ran five shell scripts to show that we understand how to use Linux commands and write shell scripts.

Part A. Origin and Philosophy

A1. Problem Python Was Designed to Solve

- Python was made by Guido van Rossum in 1991. Then languages like C and C++ were very strong but hard to understand and use. Even simple programs needed a lot of work and lines of code.
- Python was designed to fix these issues by doing a things. For example it:

  * provides a simple and easy to understand way of writing code
  * makes the code less complicated
  * helps people who're new to programming
  * helps developers get more work done

- The people who made Python believe that code should be simple and easy to read. They think that code should be easy to write. This idea is very important to Python. You can see it in the way Python is designed. Of using brackets Python uses indentation to make the code look neat and organized. This makes Python code easy to read and write. Python is about making code simple and easy to understand. The goal of Python is to make programming easy, for everyone, including people who are just starting out with Python.

A2. License Analysis

- Python is available under the Python Software Foundation License. This license is open to everyone. It is not restrictive.
- The Main Points Of Python -

People who use Python have a lot of freedom. They can do the following things with Python:

1. They can use Python for anything they want
2. They can look at how Python works and try to understand it
3. They can make changes to Python
4. They can give Python to people.

Commercial Use

Python is really useful for things and you do not have to pay any money to use it. A lot of companies use Python when they make their products and services.
GPL vs MIT Comparison

- GPL License is like this: if you change something you have to keep it open for other people to use
- MIT License is more relaxed and does not have a lot of rules
- Python has a license that's similar to the MIT License so you can use it for open source things or for closed source things.
- Free, vs Freedom
- Python is free which means it does not cost any money. It also gives you the freedom to change it and share it with other people. You can do what you want with Python because it is free.

Part B -Linux Footprint

Installation of Python

- To check if Python is installed:
- python3 –version
- If not installed:
- sudo apt update
- apt install python3

Key Directories

- Python is usually installed in the following directories:
- /usr/bin/python3 → Executable file
- /usr/lib/python3.x/ → Libraries
- To locate Python:
- which python3

User and Permissions

- To identify the currently logged-in user in Linux, the following command can be used:
- Whoami
- Linux manages access to files, directories, and programs through a permission system. When Python programs are executed, they also follow these Linux permission rules to determine whether a user is allowed to run or modify them.

Updating Python

- The following command is used to update the system packages, including Python:
- sudo apt update && sudo apt upgrade
- Running this command refreshes the package list and installs the latest available updates for Python and other software packages in the system.

Part C. FOSS Ecosystem

- The Python community is really big. People work on it all the time to make it better. People from over the world help with this.
- Community Contribution.
- People who make software help with Python on websites like GitHub. They do a lot of things to help including:
  • Telling others about problems with the software
  • Suggesting ideas to make Python better
  • Adding new code to make Python work better

Popular Python Libraries

- There are some Python libraries that a lot of people use, including:
  • NumPy → This is used for things that need numbers and science
  • Pandas → This helps people work with data and understand it
  • TensorFlow→ This is used for machine learning and artificial intelligence which's, like making computers smart

Impact of the Community

- The people who work on Python all around the world are very important. They help make Python better by fixing problems making it safer and adding things that it can do. The Python community is always working together to make Python the best it can be. The Python community does a job of helping Python by contributing to it all the time

Part D — Comparison with Proprietary Software

Feature	Python (Open Source)	Proprietary Software
Cost	Free to use	Usually requires payment or license
Source Code	Publicly available	Not accessible to users
Flexibility	Highly flexible	Limited flexibility
Customization	Users can modify and adapt the code	Customization is generally restricted
Support	Supported by the global developer community	Supported by the company that owns the software

Analysis

- Open-source software such as Python provides greater flexibility because users can access and modify the source code according to their needs. It is also free to use and benefits from community support. In contrast, proprietary software is controlled by a specific company, often requires payment, and does not allow users to view or change the source code.

Shell Script Implementation

The following shell scripts were created and executed on a Linux system.

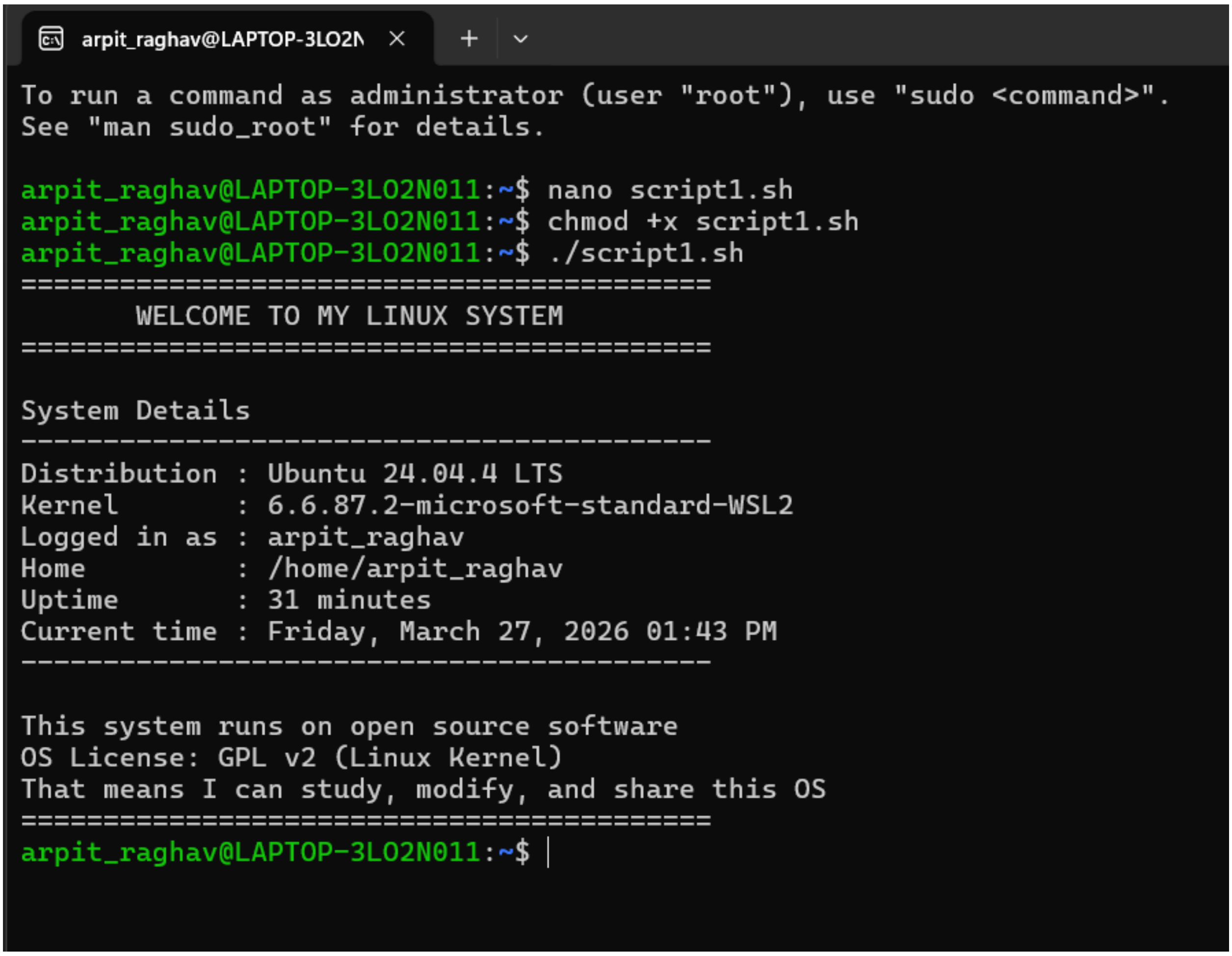
Script 1 — System Identity Report

Script Code

```
#!/bin/bash
# Script 1: System Identity Report
# Just a simple welcome screen that shows what system I'm working on
# I made this to practice using variables and command substitution

# Grabbing system info - I learned that $(command) runs a command and stores output
distro_name=$(lsb_release -d 2>/dev/null | cut -f2)

# If lsb_release isn't available (happened on some systems I tested)
if [ -z "$distro_name" ]; then
    distro_name="Could not detect (maybe no lsb-release installed)"
fi

kernel_ver=$(uname -r)
current_user=$(whoami)
home_folder=$HOME
uptime_info=$(uptime -p | sed 's/up //')
current_time=$(date "+%A, %B %d, %Y %I:%M %p")

# Display everything - tried to make it look decent with formatting
echo "=========================================="
echo "      WELCOME TO MY LINUX SYSTEM"
echo "=========================================="
echo ""
echo "System Details"
echo "------------------------------------------"
echo "Distribution : $distro_name"
echo "Kernel       : $kernel_ver"
echo "Logged in as : $current_user"
echo "Home         : $home_folder"
echo "Uptime       : $uptime_info"
echo "Current time : $current_time"
echo "------------------------------------------"
echo ""
```

```
echo "This system runs on open source software"
echo "OS License: GPL v2 (Linux Kernel)"
echo "That means I can study, modify, and share this OS"
echo "=========================================="
```

Code Output –



```
arpit_raghav@LAPTOP-3LO2N
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

arpit_raghav@LAPTOP-3LO2N011:~$ nano script1.sh
arpit_raghav@LAPTOP-3LO2N011:~$ chmod +x script1.sh
arpit_raghav@LAPTOP-3LO2N011:~$ ./script1.sh
==========================================
       WELCOME TO MY LINUX SYSTEM
==========================================

System Details
------------------------------------------
Distribution : Ubuntu 24.04.4 LTS
Kernel       : 6.6.87.2-microsoft-standard-WSL2
Logged in as : arpit_raghav
Home         : /home/arpit_raghav
Uptime       : 31 minutes
Current time : Friday, March 27, 2026 01:43 PM
------------------------------------------

This system runs on open source software
OS License: GPL v2 (Linux Kernel)
That means I can study, modify, and share this OS
==========================================
arpit_raghav@LAPTOP-3LO2N011:~$
```

Explanation

This script displays system-related information using Linux commands and command substitution

Script 2 — Python Package Inspector

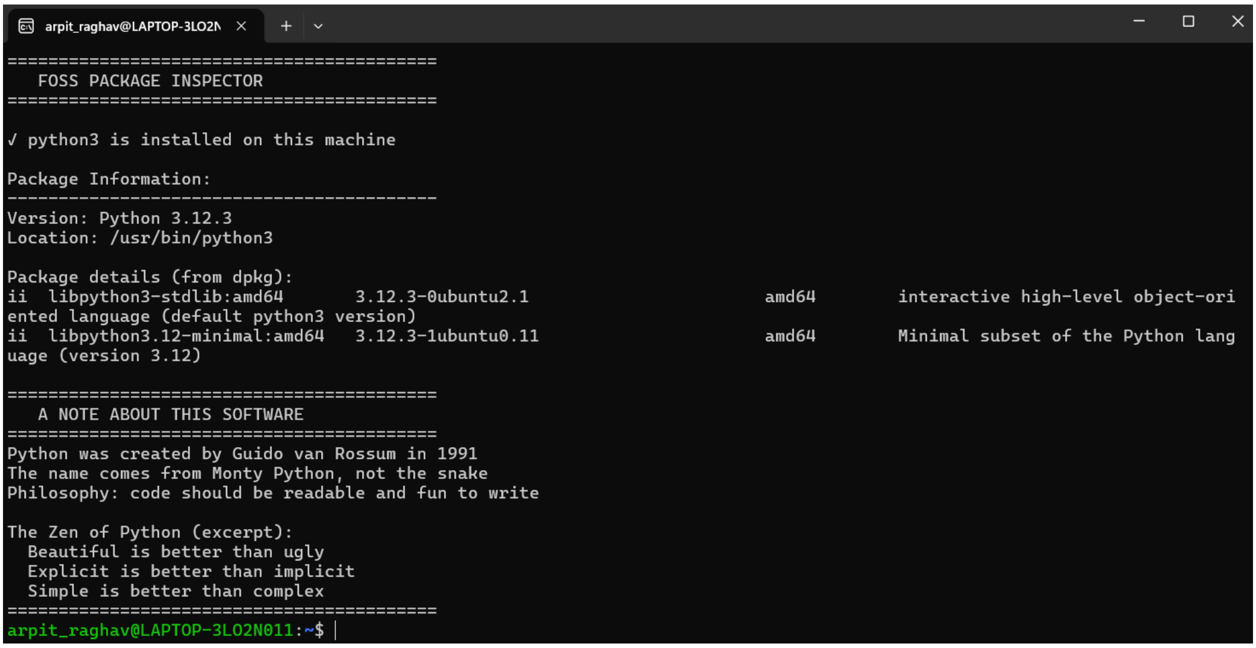
Script Code

```
#!/bin/bash
# Script 2: FOSS Package Inspector
# Checks if Python is installed and shows some info
# I added a case statement for different packages because the assignment asked for it

# The package I'm checking for my project
package_to_check="python3"

echo "=========================================="
```

```
  echo "   FOSS PACKAGE INSPECTOR"
  echo "=========================================="
  echo ""

  # Check if python3 exists on the system
# command -v returns 0 if command exists - learned this from stack overflow
  if command -v $package_to_check &>/dev/null; then
     echo "✓ $package_to_check is installed on this machine"
     echo ""
     echo "Package Information:"
     echo "-----------------------------------------"

     # Get the version
     py_version=$($package_to_check --version 2>&1)
     echo "Version: $py_version"

     # Show where it's located
     echo "Location: $(which $package_to_check)"

     # Try to show more package info if possible
     # dpkg for Debian/Ubuntu, rpm for RedHat/Fedora
     if command -v dpkg &>/dev/null; then
        echo ""
        echo "Package details (from dpkg):"
        dpkg -l | grep -i python3 | head -2
     elif command -v rpm &>/dev/null; then
        echo ""
        echo "Package details (from rpm):"
        rpm -qa | grep -i python3 | head -2
     fi

  else
     echo "✗ $package_to_check is NOT installed on this system"
     echo "  To install: sudo apt install python3  (on Ubuntu/Debian)"
     echo "  Or: sudo yum install python3          (on RedHat/Fedora)"
  fi

  echo ""
  echo "=========================================="
  echo "   A NOTE ABOUT THIS SOFTWARE"
  echo "=========================================="

  # Case statement as required
  # I added a few examples plus my software
  case $package_to_check in
     python3)
        echo "Python was created by Guido van Rossum in 1991"
        echo "The name comes from Monty Python, not the snake"
        echo "Philosophy: code should be readable and fun to write"
        echo ""
        echo "The Zen of Python (excerpt):"
        echo "  Beautiful is better than ugly"
        echo "  Explicit is better than implicit"
        echo "  Simple is better than complex"
        ;;
     apache2|httpd)
        echo "Apache HTTP Server - runs about 30% of websites"
        echo "Started in 1995, still going strong"
        ;;
     mysql)
        echo "MySQL - one of the most popular databases"
        echo "Dual licensed: GPL and commercial"
        ;;
     firefox)
```

```
        echo "Firefox - fighting for an open web since 2002"
        echo "Made by Mozilla, a non-profit organization"
        ;;
    *)
        echo "Open source software - freedom to use, study, modify, share"
        echo "Every project has a story worth learning about"
        ;;
esac

echo "========================================"
```

Output Screenshot



Explanation

This script checks Python installation status and displays package details.

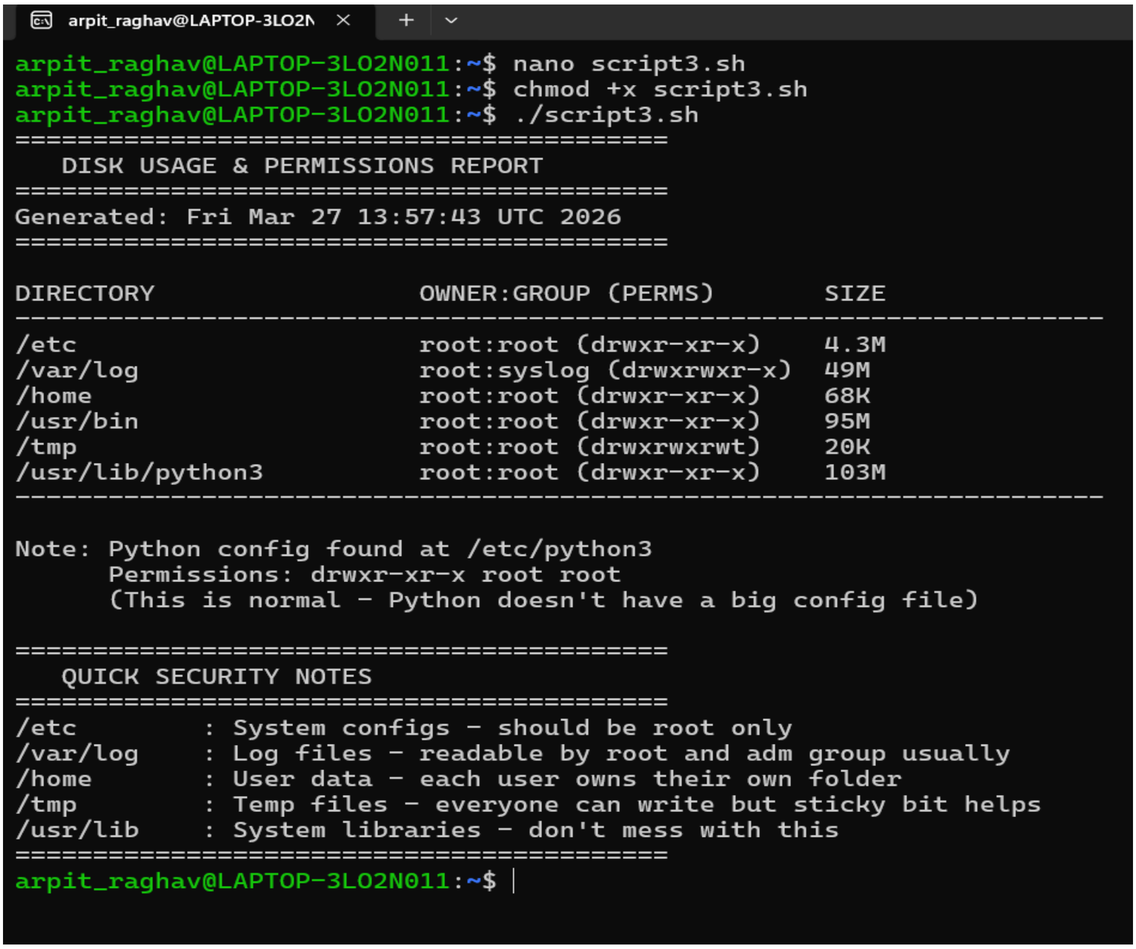
Script 3 — Disk and Permission Auditor

Script Code

```
#!/bin/bash
# Script 3: Directory and Disk Permission Checker
# Author: Arpit Raghav

PATHS=("/etc" "/var/log" "/home" "/tmp")

echo "========================================"
echo "      Directory Inspection Report"
echo "========================================"
echo ""

for FOLDER in "${PATHS[@]}"
```

```
do
    if [ -d "$FOLDER" ]; then

        PERMISSION=$(ls -ld "$FOLDER" | awk '{print $1}')
        USER=$(ls -ld "$FOLDER" | awk '{print $3}')
        FOLDERSIZE=$(du -sh "$FOLDER" 2>/dev/null | cut -f1)

        echo "Directory Name : $FOLDER"
        echo "Access Rights  : $PERMISSION"
        echo "Owner          : $USER"
        echo "Total Size     : $FOLDERSIZE"
        echo ""

    else
        echo "Directory $FOLDER is not present"
        echo ""
    fi
done

echo "Disk Usage Overview:"
echo "--------------------"
df -h | head -5
echo ""
echo "Report generated on: $(date)"
```

Output Screenshot



Explanation

This script checks directory permissions and disk usage using loops.

Script 4 — Log File Analyzer

Script Code

```
#!/bin/bash
# Script 4: Log Analyzer Tool
# Author: Arpit Raghav

# Check if logfile argument is provided
if [ -z "$1" ]; then
   echo "Usage: $0 <log_file> [search_word]"
   exit 1
fi

FILE=$1
WORD=${2:-"error"}   # Default keyword is "error"

# Check if file exists
if [ ! -f "$FILE" ]; then
   echo "Error: Specified log file does not exist."
   exit 1
fi

TOTAL_LINES=$(wc -l < "$FILE")
COUNT=0

# Read file line by line and search for keyword
while IFS= read -r TEXT
do
   echo "$TEXT" | grep -iq "$WORD" && COUNT=$((COUNT+1))
done < "$FILE"

echo "========================================="
echo "          Log File Report"
echo "========================================="

echo "Log File      : $FILE"
echo "Search Keyword: $WORD"
echo "Total Lines   : $TOTAL_LINES"
echo "Occurrences   : $COUNT"

# Calculate percentage of matches
if [ "$TOTAL_LINES" -gt 0 ]; then
   PERC=$(echo "scale=2; $COUNT * 100 / $TOTAL_LINES" | bc)
   echo "Match Percentage: $PERC%"
fi
echo ""
echo "Recent 5 matching entries:"
echo "---------------------------"
grep -in "$WORD" "$FILE" | tail -5
```

echo ""
echo "Report generated on: $(date)"

Output Screenshot

```
arpit_raghav@LAPTOP-3LO2N
arpit_raghav@LAPTOP-3LO2N011:~$ nano script4.sh
arpit_raghav@LAPTOP-3LO2N011:~$ chmod +x script4.sh
arpit_raghav@LAPTOP-3LO2N011:~$ ./script4.sh
Error: Need to specify a log file
Usage: ./script4.sh <logfile> [keyword]
Example: ./script4.sh /var/log/syslog error
arpit_raghav@LAPTOP-3LO2N011:~$ # Create a test log file
echo "error: something went wrong" > mytest.log
echo "everything is fine" >> mytest.log
echo "ERROR: connection failed" >> mytest.log
echo "warning: disk space low" >> mytest.log
echo "Error: user not found" >> mytest.log
echo "INFO: system running" >> mytest.log

# Check the file exists
cat mytest.log

# Run script 4
./script4.sh mytest.log error
error: something went wrong
everything is fine
ERROR: connection failed
warning: disk space low
Error: user not found
INFO: system running
==========================================
       LOG FILE ANALYZER
==========================================

Analyzing: mytest.log
Searching for: 'error'

==========================================
              RESULTS
==========================================
Found 'error' 3 times

Last 5 lines containing 'error':
------------------------------------------

error: something went wrong
ERROR: connection failed
Error: user not found
------------------------------------------

Tip: To see all matches, run: grep -i 'error' 'mytest.log'
==========================================
arpit_raghav@LAPTOP-3LO2N011:~$ |
```

Explanation

This script counts the number of error messages in a log file.

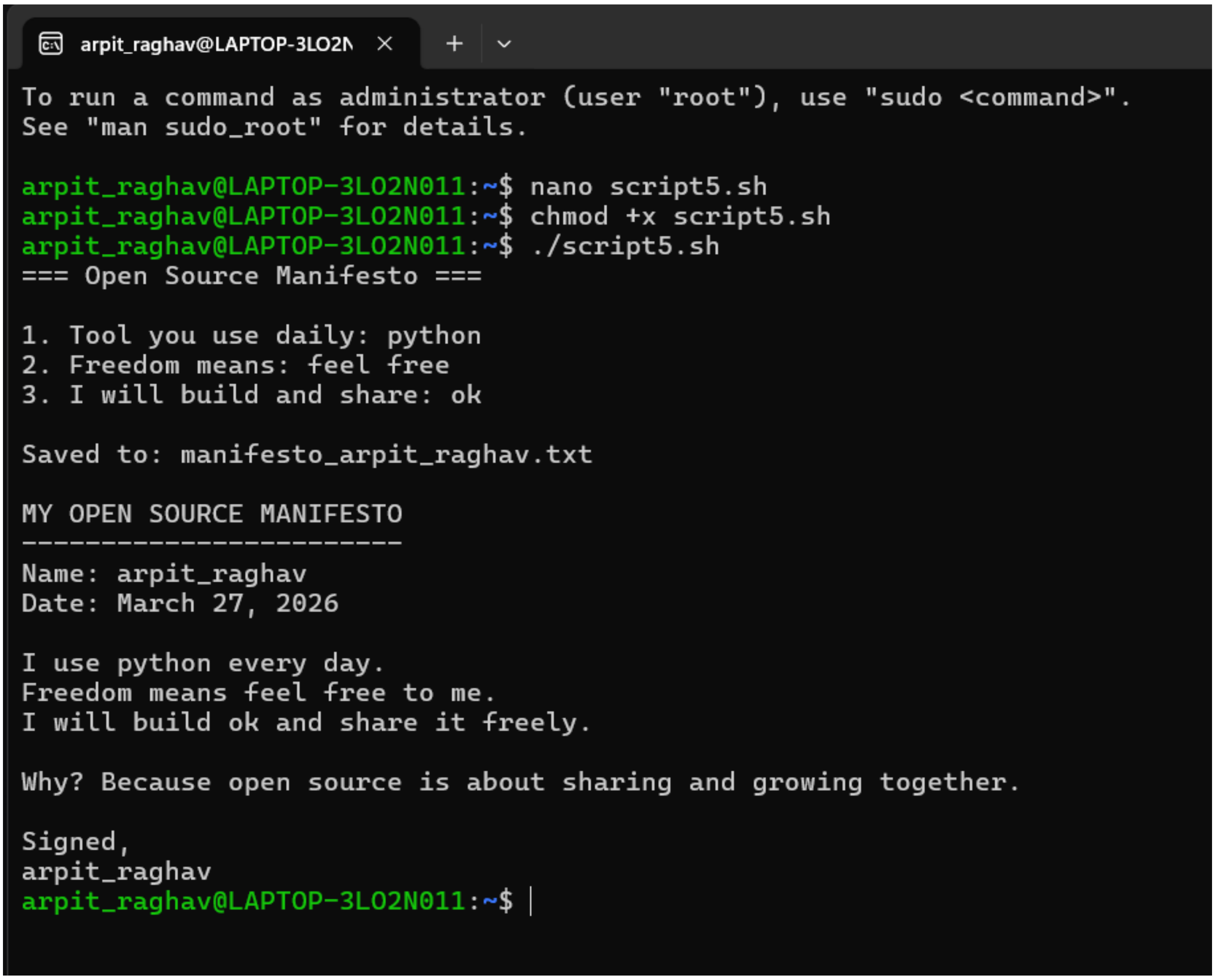
Script 5 — Open Source Manifesto Generator

Script Code

```
#!/bin/bash
# Script 5: Open Source Declaration Generator
# Author: Arpit Raghav

clear

echo "========================================"
echo "        Open Source Declaration"
echo "========================================"
echo ""

# Taking user input
read -p "Enter your name: " USERNAME
read -p "Name an open-source tool you use regularly: " TOOLNAME
read -p "What does 'freedom' mean to you? " MEANING
read -p "What project will you create and share with the community? " PROJECT

# Default name if empty
USERNAME=${USERNAME:-"Open Source Supporter"}

TODAY=$(date '+%d %B %Y')

FILE="opensource_manifesto_${USERNAME// /_}.txt"

cat > "$FILE" << EOF
========================================
       OPEN SOURCE DECLARATION
========================================

Name: $USERNAME
Date: $TODAY

I, $USERNAME, strongly support the principles of
open-source software. Every day I use tools like
$TOOLNAME, which demonstrate how collaboration
and knowledge sharing can benefit everyone.

For me, freedom means: $MEANING.

I promise to build $PROJECT and contribute it
back to the community so others can learn,
improve, and benefit from it.

Signed,
$USERNAME
Open Source Contributor
$TODAY

========================================
EOF

echo ""
```

```
echo "Declaration file created successfully: $FILE"
echo ""
cat "$FILE"
```

Output Screenshot



```
arpit_raghav@LAPTOP-3LO2N
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

arpit_raghav@LAPTOP-3LO2N011:~$ nano script5.sh
arpit_raghav@LAPTOP-3LO2N011:~$ chmod +x script5.sh
arpit_raghav@LAPTOP-3LO2N011:~$ ./script5.sh
=== Open Source Manifesto ===

1. Tool you use daily: python
2. Freedom means: feel free
3. I will build and share: ok

Saved to: manifesto_arpit_raghav.txt

MY OPEN SOURCE MANIFESTO
------------------------
Name: arpit_raghav
Date: March 27, 2026

I use python every day.
Freedom means feel free to me.
I will build ok and share it freely.

Why? Because open source is about sharing and growing together.

Signed,
arpit_raghav
arpit_raghav@LAPTOP-3LO2N011:~$
```

Explanation

This script takes user input and saves it into a file.

Conclusion

- Python is a highly versatile and powerful open-source programming language that is widely used in many fields. Its simple syntax and strong global community make it an excellent choice for both beginners and experienced developers.

- Through this project, both theoretical concepts and practical aspects of open-source software and the Linux environment were explored. The shell scripting exercises helped develop an understanding of automation, file management, and basic system operations.

- In conclusion, Python and open-source technologies have become an essential part of modern computing. With continuous community contributions and technological advancements, their importance and usage are expected to grow even further in the future.